AI CLUSTERS

Unified Full-Stack Telemetry for AI Systems

From silicon to cluster: enabling unified observability across in-band and out-of-band telemetry

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Context and What We Will Cover

Why now

AI clusters are scaling faster than the tooling that observes them.

Telemetry already exists at every layer — CPU, GPU, network, storage, platform management and orchestration.

Standards solve access and collection; correlating across layers is still manual work.

For OCP AI Clusters, hyperscalers, CSPs and the OpenTelemetry community.

What we will cover

01 **The problem —** telemetry fragmented across domains

02 **Why today's approaches fall short —** collection without correlation

03 **The missing bridge —** identity, time and semantics

04 **The proposal —** a normalization contract for cross-layer observability

05 **The ask —** what to standardize next, and contributing through the OCP AI Cluster WG, alongside Hardware Management and AI Systems

Full-Stack Telemetry – Current State & Goals

What is Full-Stack Telemetry?

- Telemetry spans CPU, GPU, NIC and fabric, storage, platform management and orchestration.
- Two collection planes exist today: in-band telemetry and out-of-band telemetry.
- Both can provide streaming, event-driven, and bulk telemetry export models.

Why is it Important?

- AI cluster jobs slow or stall for reasons that cross layer boundaries.
- Root-causing a slow training step requires telemetry from compute, fabric, platform and scheduler domains on a single timeline.

Problem Statement – No Cross-Layer Correlation

- **No shared identity for the same device across domains**
  - *Host, BMC and orchestration each name the same GPU or NIC differently — PCIe address, serial, node or pod name*
- **No common time base across collection lanes**
  - *Independent clocks and sampling intervals make sub-millisecond events impossible to align*
- **No shared semantics for what a metric means**
  - *Same counter, different unit, scope and reset behavior across vendors and stacks*
  - *PLDM, Redfish, MCTP and OpenTelemetry solve access and collection — not correlation*

Goals

- A vendor-neutral normalization architecture: common identity, timestamp provenance and semantics across GPU, CPU, accelerators, network and fabric, storage and platform controllers.
- Preserve every existing standard, interface and ownership boundary; add only the missing agreement between them.

Telemetry Is Trapped Inside System Boundaries

Every layer observes itself. No layer can reconstruct the system event.

AI Training Job training step latency increase

context boundary

Scheduler Layer	Placement

context boundary

Fabric Layer	Congestion

context boundary

Accelerator Layer	Utilization

context boundary

CPU Layer	Power

context boundary

Platform Layer	Health

WHAT IS MISSING

No Common Identity
the same device is named differently in every domain

No Common Time Reference
independent clocks and sampling intervals

No Common Semantics
same counter name, different unit and scope

Ownership Boundaries
different teams, tools and lifecycles

Telemetry collection is solved. **Cross-layer correlation remains manual.**

Why Existing Approaches Fall Short

The problem is not ownership. It is the absence of a common correlation model.

In-Band Telemetry

- OS
- Drivers
- Runtimes
- Frameworks
- Device software

Out-of-Band Telemetry

- BMC
- Redfish
- PLDM
- Platform controllers

Why Correlation Fails

- Two Planes, Never Joined
- Correlation Only After an Incident
- Per-Domain Tools and Dashboards
- No Shared Correlation Contract

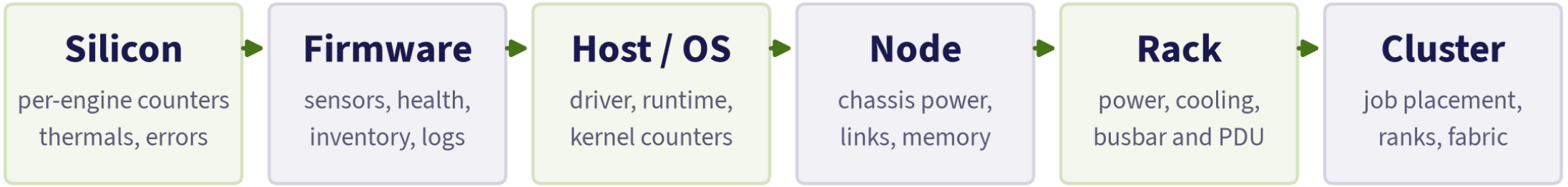
Operator Experience Today

- Manual correlation
- Multiple dashboards
- Slow root-cause analysis
- Operational complexity

Both planes work. Neither can answer a question that spans them.

Telemetry Context Gets Lost Across System Boundaries

Six scopes, five handoffs. Identity, timestamp provenance and semantics must survive every transition.



Each scope contains the ones to its left. Every arrow is a handoff — and every handoff is where telemetry loses its meaning today.

What has to survive every handoff

Common identity
one ID every scope can emit

Common time
declared clock, offset, skew

Common semantics
definition, unit, sampling method

Ownership boundaries stay where they are — each scope keeps its own producer. Only the agreement between them is new.

The Missing Bridge: Identity, Time, Semantics

Three missing agreements. Defined once here, and used by everything that follows.

Raw streams — each self-consistent, none joinable

in-band · out-of-band · bulk and event-driven

Identity

- One canonical component ID
- Emitted by every path
- Joinable by every consumer

Time

- Declared clock source
- Known offset and skew
- Comparable across lanes

Semantics

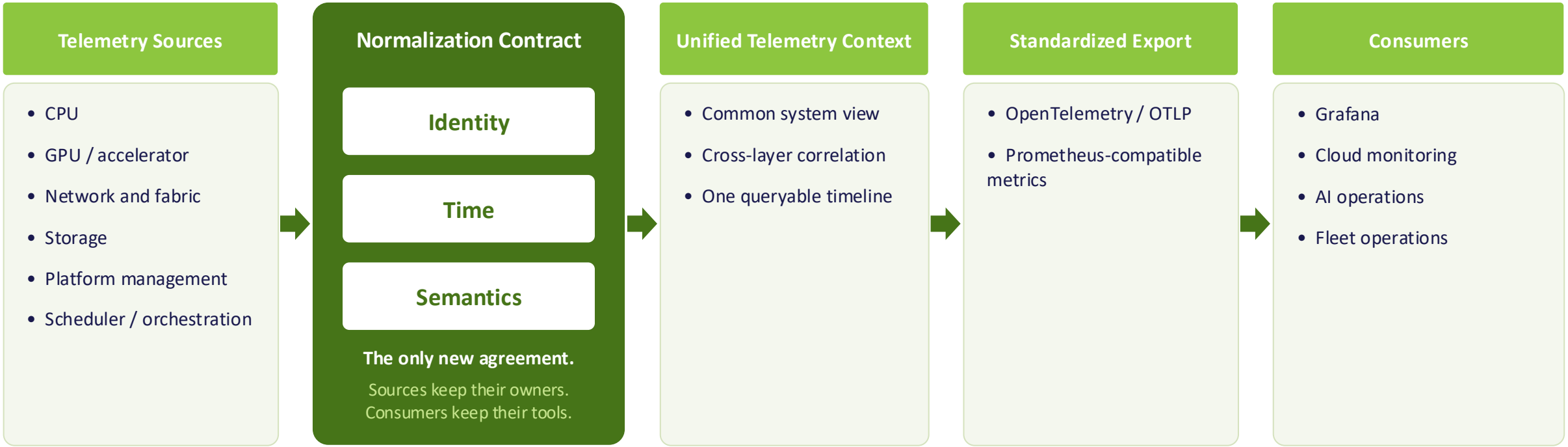
- Metric definition
- Unit
- Resource binding
- Sampling method

One queryable timeline — one incident, seen from every layer at once

OpenTelemetry hardware conventions · GPU metrics, status Development

Unified Telemetry Architecture for AI Clusters

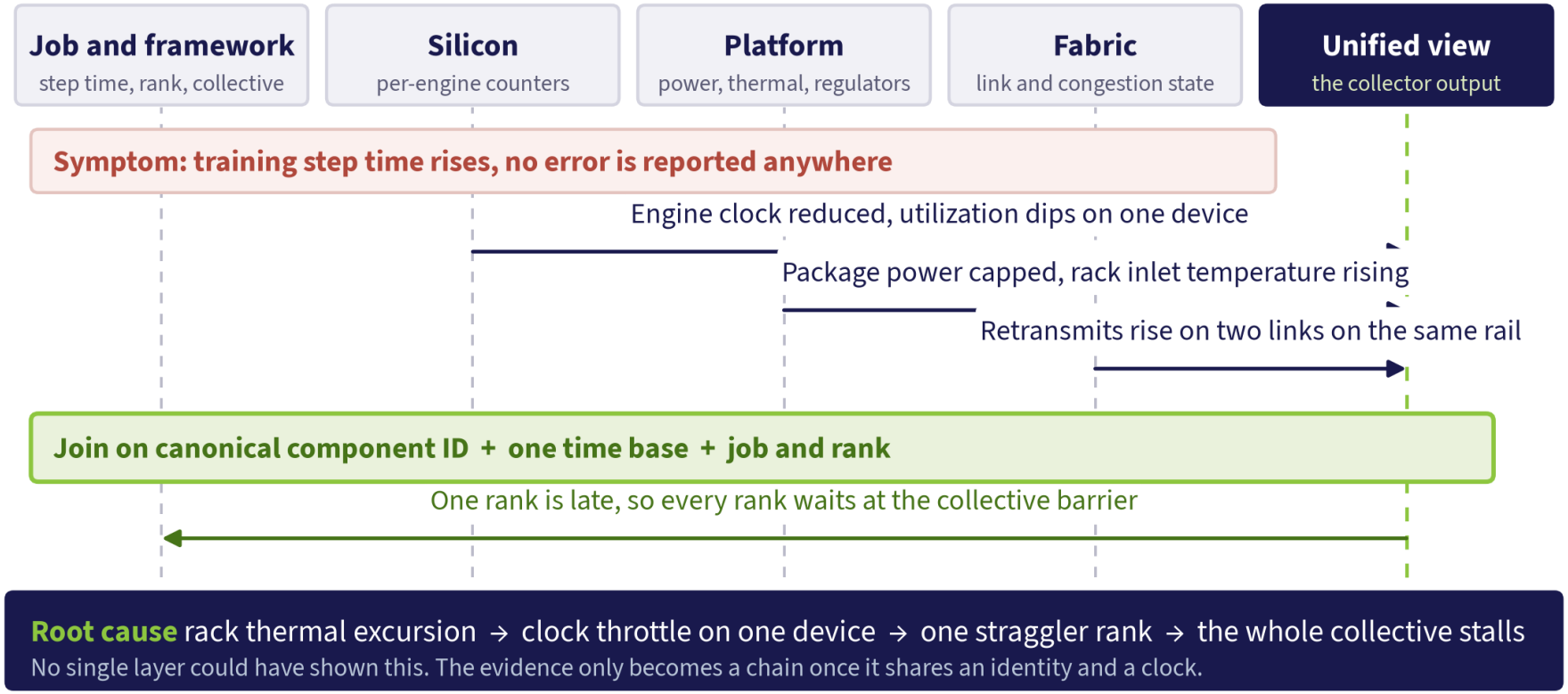
A normalization contract connects telemetry sources to a common system view.



Standardize the bridge, not the sources.

From Slow Training Step to Cross-Layer Root Cause

Symptom, signals, correlation, root cause — the join is what makes the chain causal.



Illustrative scenario, constructed to show the required data joins. No measured performance or time-to-resolution figures are claimed.

What Exists Today, and What Does Not

We are not proposing to replace anything. We are proposing to fill one empty column.

	Access and transport	Data model	Metric semantics	Cross-layer identity and one time base
Cluster and job	**covered** scheduler APIs	**covered** scheduler APIs	**partial** definitions vary	**not covered**
Fabric	**covered** Ultra Ethernet	**partial** no common model	**partial** counters, no names	**not covered**
Rack and power	**covered** Redfish	**covered** Redfish	**partial** sensor names vary	**not covered**
System and node	**covered** Redfish	**covered** Redfish	**partial** sensor names vary	**not covered**
Platform	**covered** PLDM monitoring	**covered** PLDM PDRs	**partial** IDs, no semantics	**not covered**
Silicon and accelerator	**partial** vendor specific	**partial** vendor specific	**not covered** 7 metrics, early draft	**not covered**

Transport and data models are largely solved by existing open standards, bulk snapshots included. Metric semantics are partial.

Nothing standardizes the join. That last column is the contribution we are proposing.

Source: Redfish DSP0266 · PLDM DSP0248 · PLDM DSP0242 · OCP Redfish DSP2049 · OpenTelemetry GPU conventions · Ultra Ethernet v1.0.3

Call to Action

Problem to Solve

- Agree one normalization contract: canonical component identity, timestamp provenance, published metric definitions.
- Scope it as a mapping between the standards we already have.

Community Home

- Proposed home: Scalable Cloud Infrastructure Management, within OCP AI Clusters in alignment to AI Systems and OCP Hardware Management.
- Align with Profile, RAS API, Fleet Memory Fault Management, Auxiliary Management Controller and OpenRMC.

Hardware Management · Scalable Cloud Infrastructure Management

Align, Not Replace

- DMTF PLDM DSP0248 and DSP0242, Redfish, and the public OCP manageability specifications.
- OpenTelemetry data models for normalized export; accelerator coverage there is open work.

PLDM DSP0248 · DSP0242 · OpenTelemetry hardware conventions

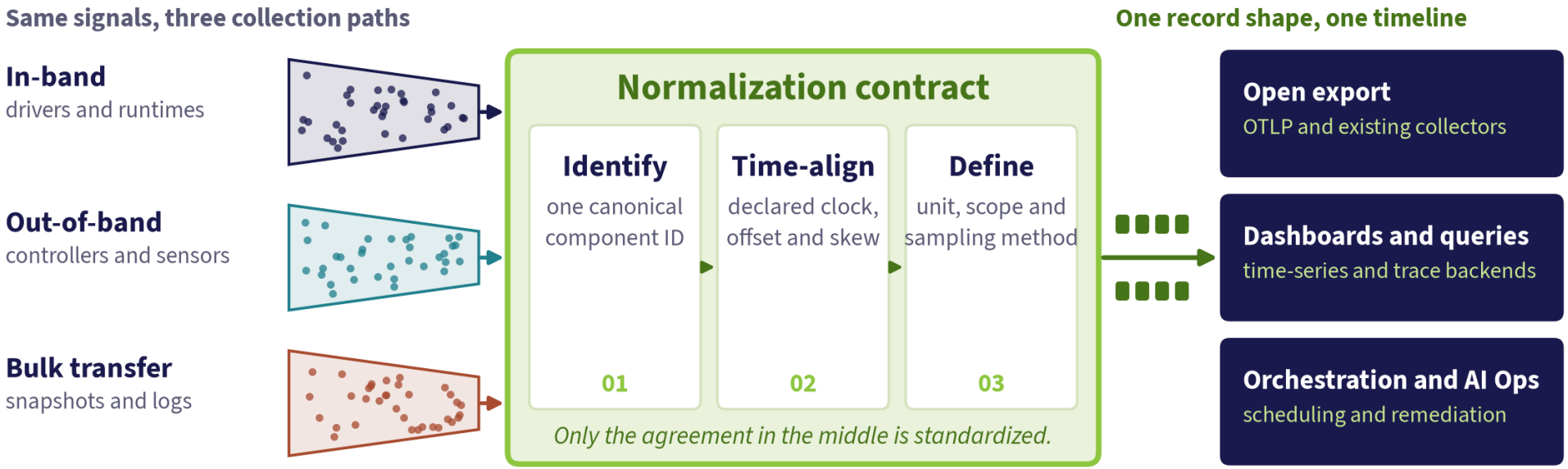
Next Step

- Bring this scope to the next Hardware Management project call; if supported, propose a draft specification outline.
- No product or facility availability is announced. Specification work only.

opencompute.org/projects · opencompute.org/products

Backup: What Normalization Does to the Data Path

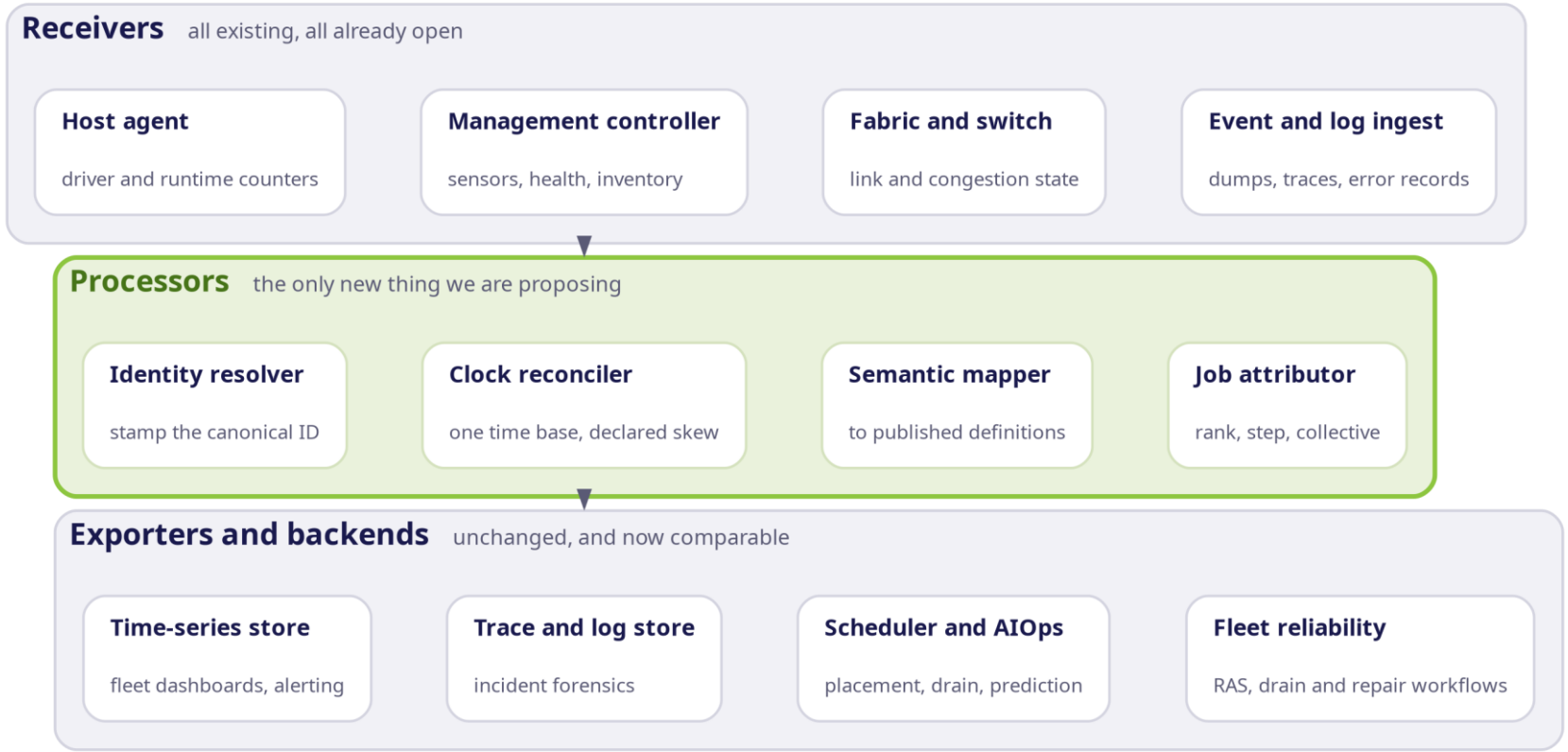
Same signals in. One identity, one clock, one definition out.



The signals do not change and the collectors do not change. What changes is that everything arrives with the same identity, the same clock and the same definition.

Backup: Reference Architecture — One Collector, One Contract

Receivers and backends already exist. The only thing that needs standardizing is the middle.



Who produces, who normalizes and who consumes at each layer is in the backup slides.

Backup: References

[OpenTelemetry — semantic conventions for hardware](#)
opentelemetry.io

[OpenTelemetry — GPU metrics conventions, status Development](#)
opentelemetry.io

[DMTF — Redfish Specification DSP0266, version 1.23.1](#)
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[DMTF — PLDM for Platform Monitoring and Control, DSP0248](#)
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[DMTF — PLDM for File Transfer, DSP0242, version 1.0.1](#)
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[DMTF — OCP Hardware Management with Redfish, DSP2049](#)
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[DMTF — Redfish telemetry streaming proposal, OCP workshop](#)
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[Ultra Ethernet — specification history, version 1.0.3](#)
ultraethernet.org

[Ultra Ethernet — progress towards the v1.0 specifications](#)
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[OCP — Hardware Management Project wiki](#)
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[OCP — Projects index](#)
opencompute.org

[OCP — Scalable Cloud Infrastructure Management](#)
opencompute.org

Every reference title is hyperlinked to the source document.